

Can a Decision Model Play a Market Participant?

Jev in the card game Figgie, compared with the rule-based traders of Ozerov et al.

Jace Yang

<https://github.com/jaceyang97/figgie-on-jev>

26 September 2026

Abstract

Jev is a decision model. It reads a description of a situation and gives a probability for each option in a list. We test whether Jev can play the part of a known type of market trader. We use Figgie, a small card game that copies a trading floor, and the rule-based traders of Ozerov, DiSilvio and Luo (2021). We use two sets of market rules: the paper’s order book (A) and the real game’s open outcry (B). The study has three stages. Stage 0 replicates the paper with rule-based traders only. Stage 1 asks Jev about 1,200 frozen game moments (4,700 requests). Stage 2 puts Jev in full games: 26 conditions, 40 games each, and each game is paired with a game in which the rule-based “twin” sits in the same seat with the same cards (112,032 requests). Main results: (1) In the paper’s 240-second setting, the rule-based traders rank as in the paper (fundamentalist, then bottom-feeder, then noise trader); in the paper’s 10,000-event setting, the bottom-feeder beats the fundamentalist in 2 of 4 line-ups. (2) Jev’s belief about the goal suit is worse than the card-counting calculation and than a uniform guess, and Jev is most wrong when it is most sure. (3) A persona text moves Jev’s actions toward the twin in 22 of 24 cells, but the similarity stays low (0.04–0.57). (4) In full games, Jev earns less than its twin: on average -34 chips per game in A and -58 in B (38 paired games per condition). No condition is clearly better than its twin; 8 of 13 conditions in A and 2 of 13 in B are not clearly different from it. The wording does not change profit, and more similar decisions do not give more similar profit.

1 Introduction

Many people now ask whether AI models can stand in for people in economic experiments. For chat models, this idea has a name: *homo silicus* (Horton, 2023). Jev is a different kind of model. It does not write text. It reads a state and a list of options, and it returns a probability for each option. This makes it easy to use as a trader in a simulated market: at each moment, the options are the legal actions.

We ask a simple question: *if we tell Jev which type of trader to be, does it act like that trader, and does it earn what that trader earns?* To answer it, we need traders whose behaviour is exactly known. The Figgie study of Ozerov, DiSilvio and Luo (2021) gives four such traders, with full algorithms. We add two market makers from the literature. Each rule-based trader is a *twin*: the reference that Jev must copy.

We have four research questions. We wrote them, and our predictions, before we collected the data.

- **RQ1, belief.** How well does Jev estimate the goal suit? H1a: with a full, factual description of the game, Jev’s estimates are not worse than the card-counting calculation. H1b: if we give Jev the result of the card-counting calculation, Jev becomes too confident. H1c: Jev’s estimates become more correct later in the game.
- **RQ2, market rules.** How do rule sets A and B change the results? H2a (replication): in A, the rule-based traders rank as in the paper (fundamentalist, bottom-feeder, noise trader). H2b: in B, the paper’s traders trade less and the fundamentalist has a smaller advantage. H2c: the profit difference between Jev and its twin is different in A and B (no direction predicted).
- **RQ3, persona wording.** Can a text description control Jev’s actions? Which type of description works better? H3a: Jev obeys simple yes-or-no rules (for example “never take”) with both wordings. H3b: Jev does not do calculations that a description tells it to do. H3c: a behaviour description can

change Jev’s actions in ways that it does not state (Jev reads the words literally). H3d: the effect of the wording is the same in A and B.

- **RQ4, profit.** Does Jev make more or less money than its twin in the same seat? H4a: Jev without a persona makes less than the fundamentalist but more than the noise trader. H4b: the more similar Jev’s decisions are to a rule-based trader, the more similar its profit is to that trader’s.

We fixed three primary measurements in advance: the change in similarity to the twin that each persona causes (decision layer), the difference in log loss between Jev and card counting (belief), and the mean profit difference between Jev and its twin in each rule set (game layer). All other results are secondary.

2 The game and the model

Figgie. The deck has 40 cards in four suits. One suit has 12 cards, one has 8 and two have 10. No player knows which suit has how many. The *goal suit* is the suit of the same colour as the 12-card suit; it has 8 or 10 cards. Each of four players starts with 350 chips, pays 50 into a pot of 200 and gets 10 cards. For 4 minutes, the players buy and sell cards for chips. At the end, each goal-suit card pays 10 chips from the pot, and the player with the most goal-suit cards gets the rest of the pot (100 or 120 chips; a tie splits it). Profit is final chips minus starting chips (350). The game is zero sum.

Jev. Jev (`typesafe/jev-1.13`, served as `typesafe/jev-1.13-20260917`) reads a JSON state of up to 64,000 tokens and answers one or more questions. A *choice* question has up to 255 options, each with a one-line description. Jev returns a probability for every option. We call it through OpenRouter. Input costs US\$0.042 per million tokens; output is free.

Technical details

There are 12 possible decks (3 choices of the 12-card suit’s partner position $\times$ 4 suits, Table 1 of the paper). A request is `{model, state, questions}`. Each question has a `type`, an instruction text and, for `choice`, a map from option label to description. The answer holds `choice` (the most probable label) and `probabilities` (all labels). We never use Jev’s own `choice` field; we sample (section 3.3).

3 Method

3.1 The simulator

The simulator is event driven, as in the paper. Each trader wakes after a random gap (exponential, mean $1/\lambda$). It looks at the market and decides. If it sends an order, the order reaches the market Λ seconds later, and the trader wakes again only after that. A game lasts 240 seconds, or, in the paper’s setting, 10,000 events.

Two rule sets.

- **A, order book** (the paper’s Algorithm 1). Many orders can wait at each price. A new order trades with the best waiting order on the other side, at the waiting order’s price. Other orders stay after a trade. A player keeps at most 5 orders per side and suit (the oldest goes). A player can cancel its orders in a suit.
- **B, open outcry** (the real game). Each suit has one best bid and one best ask. A new bid must be higher than the best bid; a new ask must be lower than the best ask. Every trade removes all bids and asks in all suits.

Speed. In stages 1 and 2, all traders, Jev included, wake 0.5 times per second on average and their orders arrive 0.3 seconds later (the *equal speed*). So a profit difference cannot come from speed. For the replication, we also use the paper’s speed (1 wake per second, no delay).

Technical details

Latency. The paper sends an order 2Λ after the moment the trader sees the market (its method (b)). Our parameter is this total delay, so the paper’s slow traders ($\Lambda = 100$) have a delay of 200. **Order book details.** Price–time priority; self-trade prevention (a trader’s own opposite order is removed, not filled); an order whose owner can no longer pay or deliver is removed when it reaches the top. The paper trades at the buy order’s price; we trade at the waiting order’s price, which is the standard rule in real order books. **Common random numbers.** Game g has seed $s(g)$. The seed fixes the deal and one random stream per seat for wake-up gaps and for the trader’s own random choices. So game g has the same cards, in the same seats, in every condition, and the opponents’ streams are the same in the Jev arm and in the twin arm.

3.2 The rule-based traders

All four traders of the paper send orders with the same rule (the paper’s Algorithm 2): pick buy or sell with equal chance; to buy, draw a price between 0 and the buy value; to sell, draw a price between the sell value and twice the sell value; take the standing price if it is better. The traders differ only in how they value a suit.

- **Fundamentalist.** Values a suit from *card counting*: the cards it knows exist (its hand, plus cards it saw others sell), the probability of each of the 12 decks, and the payout of one more card, including its share of the majority prize. It deletes its orders that no longer agree with its values (the paper’s Algorithm 4).
- **Bottom-feeder.** Copies the fundamentalists: its value is the middle of their last 4 buy and last 4 sell prices. In stages 1 and 2, it never copies the seat under test, so the opponents act the same in every condition.
- **Chartist.** Expects the recent price trend to continue (Chiarella, Iori and Perelló, 2009).
- **Noise trader.** Values a suit at the best bid times e^Z , $Z \sim N(0, 1)$.
- **Market maker, Glosten–Milgrom (GM) (1985).** Quotes the expected card value after a buy (ask) and after a sell (bid), from card counting and the direction of other traders’ trades. Never takes another trader’s order.
- **Market maker, Avellaneda–Stoikov (AS) (2008).** Quotes around the market price, moves the quotes against the cards it holds, and quotes wider when prices move a lot and early in the game. Never takes another trader’s order.

Technical details

Fundamentalist value (paper, section 2.3): $e_b(j, n, m) = \sum_i m_i v(i, j, n)$ with $v = 0$ if j is not the goal suit of deck i , else $10 + v_m(i, n)$; $v_m(i, n) = a_i r^n$ for $n < x_i$, else 0, with $r = 1.2$, $a_i = p_i(1 - r)/(1 - r^{x_i})$, $x_i \in \{5, 6\}$ the majority size and $p_i \in \{120, 100\}$ the prize. $e_s(j, n, m) = e_b(j, n - 1, m)$. **GM:** belief $\pi_s \propto cc_s \cdot \prod$ trade-flow likelihoods with a share $\mu = 0.3$ of informed traders; ask = $10 \pi(1 + \mu)/(\pi(1 + \mu) + (1 - \pi)(1 - \mu))$, bid with $\mu \rightarrow -\mu$. **AS:** reservation price $r = \text{mid} - q\gamma\sigma^2\tau$, half spread $\gamma\sigma^2\tau/2 + \ln(1 + \gamma/k)/\gamma$, with $\gamma = 0.1$, $k = 1.5$, $q = \text{cards held minus cards dealt}$, σ^2 from trade-price changes, $\tau = \text{share of the game left}$.

3.3 The Jev trader

At each wake-up, the Jev trader sends one request with two questions: “which action?” and “which suit is the goal suit?”. The request has four parts.

1. **Rules.** A short text with the correct rules of Figgie and of the rule set. It has no strategy hints.
2. **State.** Hand, chips, time left, the order book (A: best 3 prices per side and its own orders; B: best bid and ask), and the version of the context under test: *basic* (only this), *+summary* (facts computed from the public record: trade counts and last prices per suit, each player’s net buys and sells, and Jev’s own recent actions and their results) or *+log+summary* (also the full list of trades). Two further versions add values that code calculates, and we use them only as tests: *+known* (cards known to exist) and *+assist* (the card-counting probabilities).

3. **Persona.** None (*neutral*), the *algorithm* wording (the twin’s steps, as text) or the *behaviour* wording (what the trader does, without steps or numbers).
4. **Actions.** Pass; take the best bid or ask; cancel (A only); or bid or ask at a price from a ladder of 30 prices (1–20, then 22–40 in steps of 2). This gives at most 253 options.

Jev’s action is a random draw from its probabilities (*sampling*). The earlier study took the most probable option and always chose a bid, because the bid had the most options. Sampling keeps Jev’s randomness, which the persona algorithms need.

3.4 The three stages

Stage 0: replication (free). The paper’s line-ups (figures 3–5, tables 2–3), 100 games each, in three settings: A with 10,000 events, A with 240 seconds, and B with 240 seconds; each at the paper’s speed and at the equal speed. We also measure the profit noise of each trader in the stage 2 line-up.

Stage 1: decision layer (4,700 requests, US\$1.02). We play rule-based games in the stage 2 line-up and copy the tested seat’s view at random wake-ups: 100 *frozen moments* per rule set and trader type (at most 3 per game, only where the twin can act). For each moment we know the true goal suit, the card-counting probabilities and the twin’s action distribution (4,000 draws of its random choices). Then:

- **1a, persona wording** (3,600 requests): each moment with the algorithm persona, the behaviour persona and no persona, state version +summary.
- **1b, state versions and belief** (1,000 requests): the fundamentalist seat’s moments, no persona, all five state versions.
- **1c, repeatability** (100 requests): 50 moments asked twice.

Stage 2: game layer (112,032 requests, US\$28.50). One seat is Jev; the other three are a fundamentalist, a bottom-feeder and a noise trader (the paper’s line-up types; the chartist loses so much that it would give free profit to the others). There are 26 conditions: 2 rule sets $\times$ (neutral + 6 types $\times$ 2 wordings). Each condition plays 40 games, fixed in advance. Each condition has a twin arm: the rule-based trader of the same type in the same seat and games (the twin of neutral is the fundamentalist). Seats rotate from game to game.

Technical details

Similarity. For Jev’s distribution π and the twin’s τ over the menu, $\text{overlap} = \sum_a \min(\pi(a), \tau(a)) = 1 - \text{TV}(\pi, \tau)$. “Actions only” renormalises both without pass. A twin price outside the ladder moves to the nearest ladder price (3.1% of the twin’s probability mass). **Belief.** Log loss $-\ln p(\text{goal})$ (primary; uniform $= \ln 4 \approx 1.386$), Brier score $\sum_s (p_s - \mathbb{I}[s = \text{goal}])^2$ (uniform $= 0.75$), top-1 accuracy, calibration. **Game layer.** $d_g = \text{profit}_{\text{Jev}} - \text{profit}_{\text{twin}}$ in the same seat and seed. We fit $d_g = \beta_0 + \beta_A \mathbb{I}[A] + \beta_D \mathbb{I}[\text{behaviour}] + \beta_{AD} \mathbb{I}[A] \mathbb{I}[\text{behaviour}] + \sum_k \gamma_k \mathbb{I}[\text{type} = k] + \varepsilon_g$, and again with the number of goal cards in the starting hand as a covariate. **Uncertainty.** All 95% intervals come from a cluster bootstrap over games (4,000 resamples; 1,000 for the regression). In stage 1, the 24 persona-versus-neutral comparisons get a Holm correction. **Selection rule for stage 2.** Use the cheapest factual state version whose log loss is not clearly worse than the best one (paired 95% interval includes 0).

4 Results

4.1 Stage 0: replication

Figure 1 compares our rule-based games with the paper’s figures 3–5.

- **The fundamentalist wins in every line-up and every setting**, and the noise trader loses, as in the paper. The more fundamentalists in a game, the less each one earns, as in the paper.
- **The order $f > b > n$ holds in all four bottom-feeder line-ups in both 240-second settings** (A and B, both speeds). In the 10,000-event setting, the bottom-feeder earns more than the fundamentalist in two line-ups ($f+b+2n$: 360 vs 306 chips; $2f+b+n$: 152 vs 96). So hypothesis H2a holds for 240-second games only.

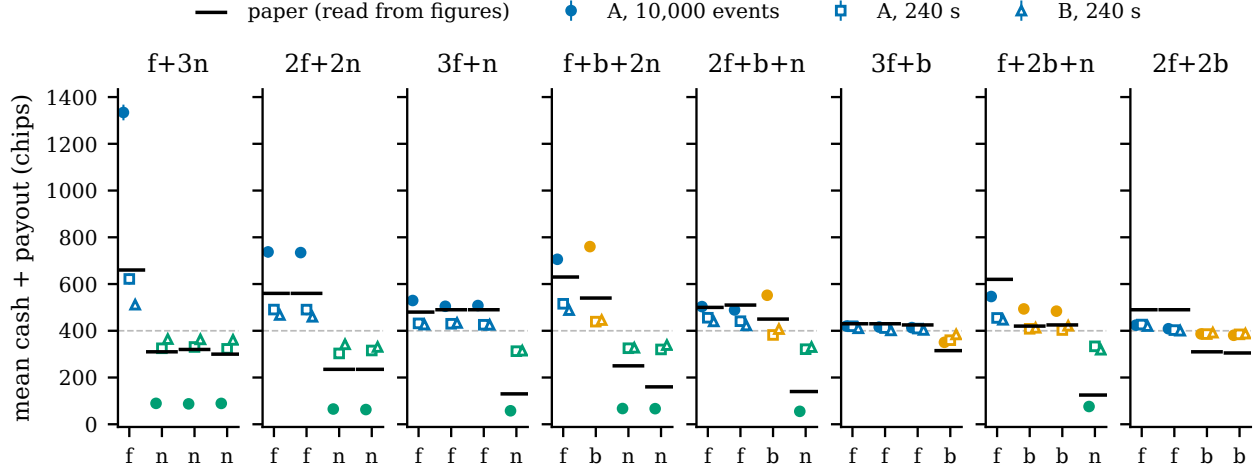


Figure 1: Mean cash + payout per seat (the paper’s measure; profit + 400), 100 games per line-up, paper’s speed. Black bars: values read from the paper’s figures (approximate). Error bars: ± 2 SE. f = fundamentalist, b = bottom-feeder, n = noise trader. The dashed line (400) is the break-even value.

- **Our profits are larger than the paper’s in the 10,000-event setting and smaller in the 240-second setting.** In 10,000 events, our noise traders lose almost all their chips (final cash about 14 of 300). The paper’s figures fall between our two settings.
- **Homogeneous games (the paper’s table 2).** Four fundamentalists: SD of final wealth, cash and payout 57.8, 30.0 and 67.8 (paper: 61.8, 33.1, 68.6). Four bottom-feeders: no trades (paper: no trades). Four noise traders: *no trades* (paper: SD of cash 188.5). The paper does not say what a noise trader does when a suit has no bid; in our reading of the paper it has no value and does not trade, so a market of noise traders never starts.
- **Latency (the paper’s table 3).** One fast fundamentalist (delay 0) against three slow ones (delay 200): each slow trader ends with more cash than the fast one (+47 to +51 chips; paper +58 to +66) and less payout (−33 to −39; paper −20 to −50). The wealth difference (+10 to +14) has the paper’s sign (+12 to +41), but its interval includes 0.
- **Rule set B** makes trade rarer (for f+3n: 35 vs 114 trades per game in A) and halves the fundamentalist’s advantage (112 vs 222 chips), as predicted (H2b).

Competitor	95% interval, slow minus fast (chips)		
	Wealth	Cash	Payout
slow0	[−8.2, 36.9]	[39.8, 54.1]	[−56.3, −8.9]
slow1	[−8.8, 33.3]	[43.5, 57.3]	[−60.2, −16.2]
slow2	[−11.4, 32.0]	[41.2, 53.9]	[−59.9, −15.0]
Paper, slow0	[15.7, 38.4]	[58.1, 65.8]	[−46.8, −23.0]

Table 1: Latency test (A, 10,000 events, 100 games), as the paper’s table 3.

Noise check for stage 2. In the stage 2 line-up at equal speed, the SD of the profit difference between two rule-based types in the same seat and games is 62–97 chips. With 40 games, this gives a minimum detectable difference of 28–43 chips per condition (Table 5, appendix).

4.2 Stage 1: decisions at frozen moments

Belief (RQ1). Figure 2 and Table 6 show the goal-suit belief.

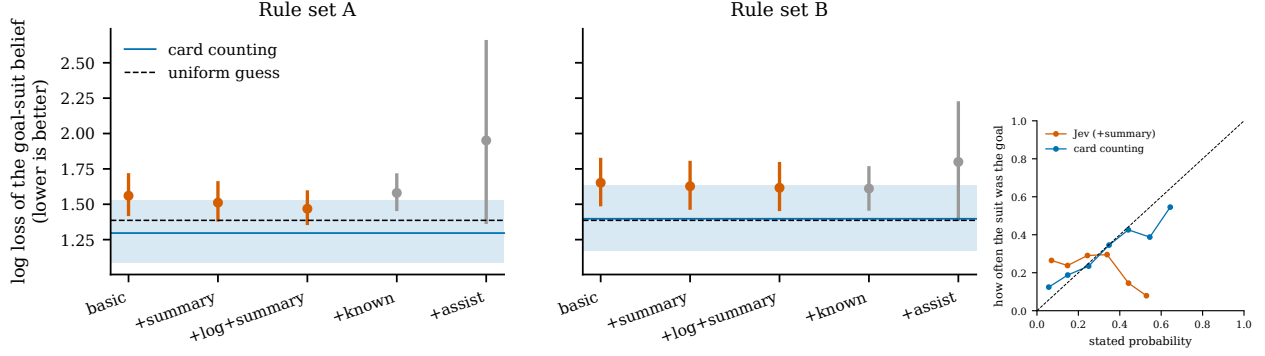


Figure 2: Left: log loss of Jev’s goal-suit belief for five state versions (100 fundamentalist moments per rule set; 95% intervals). Grey: versions with values calculated by code (tests only). Blue line and band: card counting at the same moments. Right: calibration of the +summary version, both rule sets (400 suit probabilities per curve).

- **Jev is worse than card counting** in every state version (log loss difference +0.17 to +0.28 in A and +0.21 to +0.26 in B), and **worse than a uniform guess** (1.47–1.65 vs 1.386). H1a is not supported. The best version is +log+summary in A (1.47; its interval against card counting just includes 0).
- **Card counting itself is weak:** log loss 1.30 in A and 1.40 in B, close to the uniform 1.386. The paper’s calculation uses few facts, and our moments come from all game times.
- **Jev is most wrong when it is most sure** (Figure 2). When Jev gives a suit 0.4–0.6, that suit is the goal suit only 8–15% of the time.
- **With the card-counting probabilities in its state (+assist), Jev becomes too confident** (H1b supported). Its mean top probability rises to 0.69, much higher than the 0.49 of the numbers it was given, and its log loss rises to 1.95 (A) and 1.80 (B), although its top-1 accuracy rises to the level of card counting (0.45 and 0.37).
- **Later moments are not clearly better** (H1c): with +summary, log loss falls from 1.55 to 1.48 in A and rises from 1.57 to 1.69 in B (earlier vs later half of moments).
- **State version for stage 2.** Against the best factual version (+log+summary), +summary is not clearly worse (A: +0.043, interval $[-0.006, 0.091]$; B: +0.011, $[-0.037, 0.057]$); basic is clearly worse in A (+0.092, $[0.007, 0.183]$). By the rule we fixed in advance, stage 2 uses +summary (about 5,700 tokens per request in A and 4,900 in B).

Persona wording (RQ3). Figure 3 shows the similarity between Jev and the twin.

- **A persona moves Jev toward the twin in 22 of 24 cells** (Holm-corrected). The gains are 0.01–0.21. Two cells move away: the algorithm fundamentalist in B (−0.035) and the behaviour noise trader in B (−0.063).
- **The similarity stays low.** The best cells are the noise trader (0.55, algorithm, both rule sets) and the bottom-feeder in B (0.57). The fundamentalist, the type with the most calculation, reaches only 0.36–0.43 (H3b supported: Jev does not do the calculation).
- **The market makers are almost never matched** (0.04–0.10), because their twin posts one precise price per side and Jev spreads its probability over many prices. Jev also takes other traders’ orders 1–14% of the time, although both market-maker personas say “never take” (H3a only partly supported; the rule is obeyed better in B).
- **The words can act literally** (H3c). The behaviour noise-trader text says “you take the current best bid as a loose anchor”. In B, Jev then buys with 57% of its probability, while the twin buys 8% (Figure 6).
- **Neither wording is better overall**, and the effect of the wording is different in A and B (H3d not supported): the algorithm wording gives the higher similarity for 5 of 6 types in A but only 2 of 6 in B.

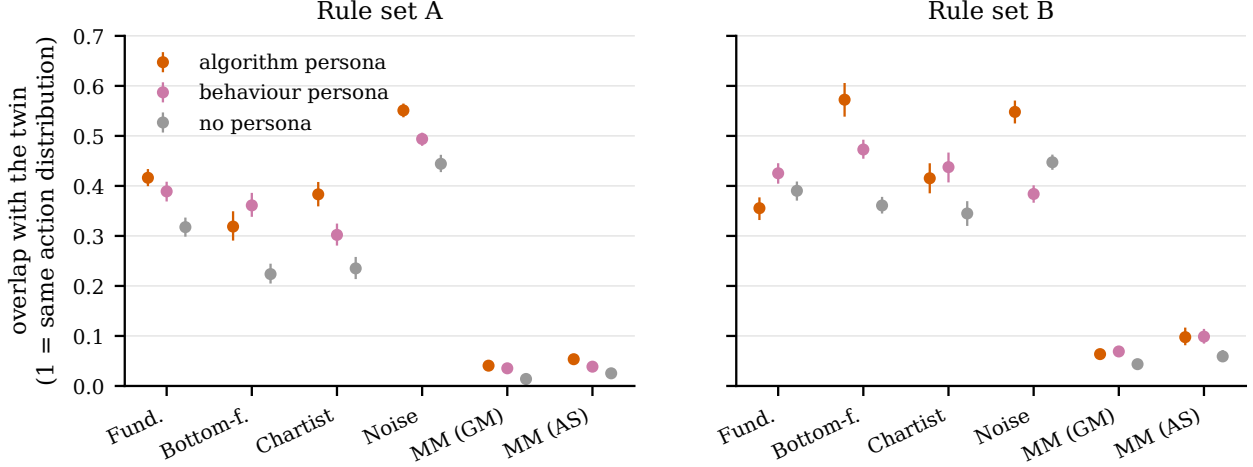


Figure 3: Similarity (overlap) between Jev’s action distribution and the twin’s, 100 moments per point, 95% intervals.

Condition	Rule set A (order book)			Rule set B (open outcry)		
	Jev	Twin	Jev – twin [95%]	Jev	Twin	Jev – twin [95%]
Neutral	−94	28	−121 [−158, −85]*	−60	14	−75 [−111, −36]*
Fund. (algorithm)	−77	28	−105 [−149, −62]*	−90	14	−105 [−149, −59]*
Fund. (behaviour)	−62	28	−90 [−126, −55]*	−70	14	−84 [−121, −46]*
Bottom-f. (algorithm)	−52	−9	−43 [−64, −20]*	−67	−11	−56 [−86, −27]*
Bottom-f. (behaviour)	1	−9	+10 [−14, 34]	−59	−11	−48 [−75, −23]*
Chartist (algorithm)	−41	−30	−11 [−34, 12]	−73	−25	−48 [−77, −18]*
Chartist (behaviour)	−60	−30	−30 [−55, −3]*	−86	−25	−61 [−95, −25]*
Noise (algorithm)	−47	−32	−15 [−46, 15]	−117	−34	−83 [−116, −45]*
Noise (behaviour)	−59	−32	−28 [−63, 8]	−119	−34	−85 [−112, −58]*
MM (GM) (algorithm)	−14	−6	−8 [−21, 7]	−57	−7	−50 [−71, −28]*
MM (GM) (behaviour)	−21	−6	−14 [−38, 7]	−55	−7	−49 [−73, −22]*
MM (AS) (algorithm)	−7	−10	+3 [−15, 22]	−44	−36	−8 [−40, 23]
MM (AS) (behaviour)	−2	−10	+8 [−12, 27]	−40	−36	−3 [−36, 25]

Table 2: Stage 2: mean profit per game in the tested seat (chips), 38 paired games per cell. *: the 95% interval does not include 0.

Repeatability. The same request gives almost the same answer: over 50 moments asked twice, the mean total variation distance is 0.071 for the action and 0.049 for the goal suit, and the most probable action is the same in 96% of pairs. So the differences above are not noise from Jev.

4.3 Stage 2: full games

In each game, one seat is Jev and the other three seats are a fundamentalist, a bottom-feeder and a noise trader. The twin arm plays the same games with the rule-based twin in Jev’s seat. We compare the two arms game by game. Each condition has 38 paired games (section 5 explains why not 40). Figure 4 and Table 2 give the result for each condition.

Profit (RQ4, primary result).

- **Jev earns less than its twin.** Over all 13 conditions of a rule set, the mean difference is −34 chips per game in A (95% interval [−44, −25]) and −58 in B ([−68, −47]).
- **No condition is clearly better than its twin.** The difference is clearly below 0 in 5 of 13 conditions in A and in 11 of 13 in B. Eight conditions in A are not clearly different from their twins: the bottom-

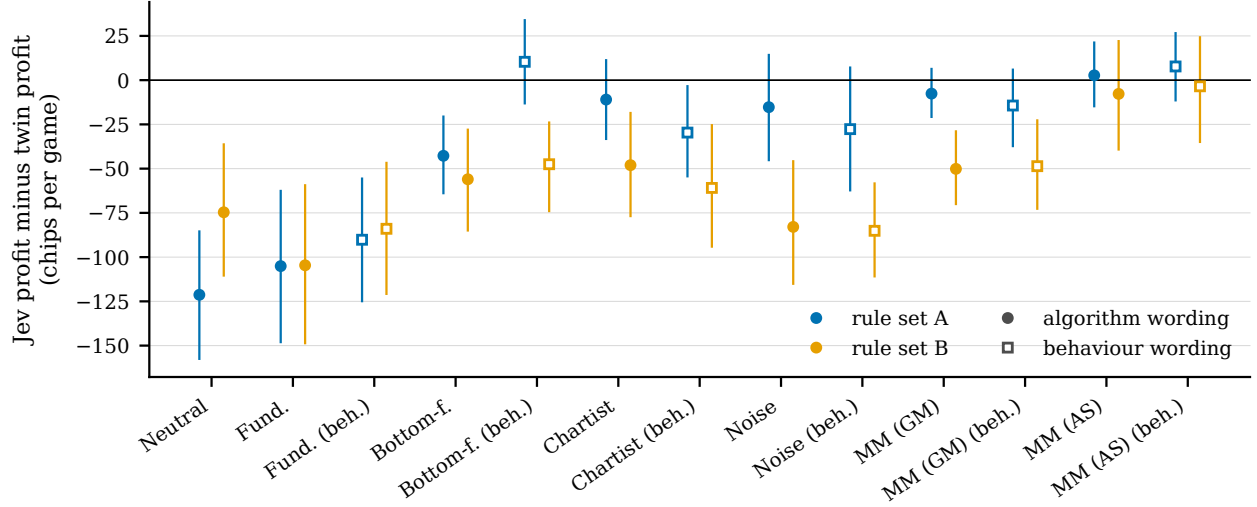


Figure 4: Stage 2: mean profit difference, Jev minus its twin, in the same seat and the same games (38 games per point; 95% cluster-bootstrap intervals). Below 0: Jev earns less than the rule-based trader it plays. The twin of “Neutral” (no persona) is the fundamentalist.

feeder with the behaviour wording, the chartist with the algorithm wording, both noise traders and all four market makers. In B, only the two AS market-maker conditions are not clearly different.

- **The largest losses are in the fundamentalist seat.** Jev with the fundamentalist persona loses 84 to 105 chips per game against the fundamentalist, in both rule sets and with both wordings. The fundamentalist is the only type that earns money here (28 chips in A, 14 in B), so it is the hardest twin to equal.
- **Close to the twin is not the same as good.** The market-maker twins earn little and Jev as a market maker loses little. A small difference in these cells shows that both traders do little, not that Jev copies the twin (their decisions overlap only 0.04–0.10 in stage 1).

Jev without a persona (H4a). The neutral Jev earns less than the fundamentalist in the same seat and games (–121 chips in A, [–158, –85]; –75 in B, [–111, –36]). It also earns less than the noise trader in A (–62, [–92, –31]); in B, the difference to the noise trader is not clear (–27, [–62, 11]). So H4a holds only in its first part: Jev with no persona is not better than a trader that decides at random.

Market rules (H2c) and wording. Jev is closer to its twin in A than in B: the difference $A - B$ is +24 chips per game ([11, 36]). So H2c is supported. The wording does not change profit: behaviour minus algorithm is +5 chips ([–6, 16]). The regression gives the same result (Table 3): the rule-set effect is +18 ([3, 31]); the wording effect (–2) and its interaction with the rule set (+13) have intervals that include 0.

The starting hand. Each extra goal card in the starting hand lowers the difference by 7 chips ([–13.8, –0.8]). A good hand helps the twin more than it helps Jev. This agrees with the belief result below: the twin uses its cards to find the goal suit, and Jev does this badly.

Decisions and similarity (H4b). Stage 1 measured, for each of the 24 persona cells, how similar Jev’s decisions are to the twin’s (overlap). If similar decisions give similar profit, cells with a higher overlap should have a smaller profit gap $|d|$. The result is the opposite: the rank correlation between overlap and $|d|$ is +0.52 (permutation $p = 0.010$, 24 cells). The market makers cause most of this: they have the lowest overlap and a small $|d|$ (see above). Inside one rule set the correlation is also positive but not clear (A: +0.55, $p = 0.07$; B: +0.27, $p = 0.41$). So H4b is not supported. At the overlap levels that Jev reaches (at most 0.57), the profit gap depends on the twin’s type, not on how well Jev copies it.

Term (reference: neutral Jev, rule set B, algorithm wording)	Estimate	95% interval
Intercept	-107	[-134, -80]
Rule set A	+18	[3, 31]
Behaviour wording	-2	[-18, 13]
Rule set A $\times$ behaviour wording	+13	[-7, 32]
Type: fundamentalist	0	[-29, 29]
Type: noise trader	+43	[9, 76]
Type: chartist	+58	[27, 88]
Type: bottom-feeder	+62	[35, 86]
Type: market maker (GM)	+66	[40, 90]
Type: market maker (AS)	+95	[58, 127]
Goal cards in the starting hand (second model)	-7.1	[-13.8, -0.8]

Table 3: Regression of the paired difference d_g on the design (988 pairs; cluster bootstrap over games, 1,000 resamples). With the goal-card covariate, all other estimates stay the same, because every condition plays the same deals: the covariate is not correlated with the design.

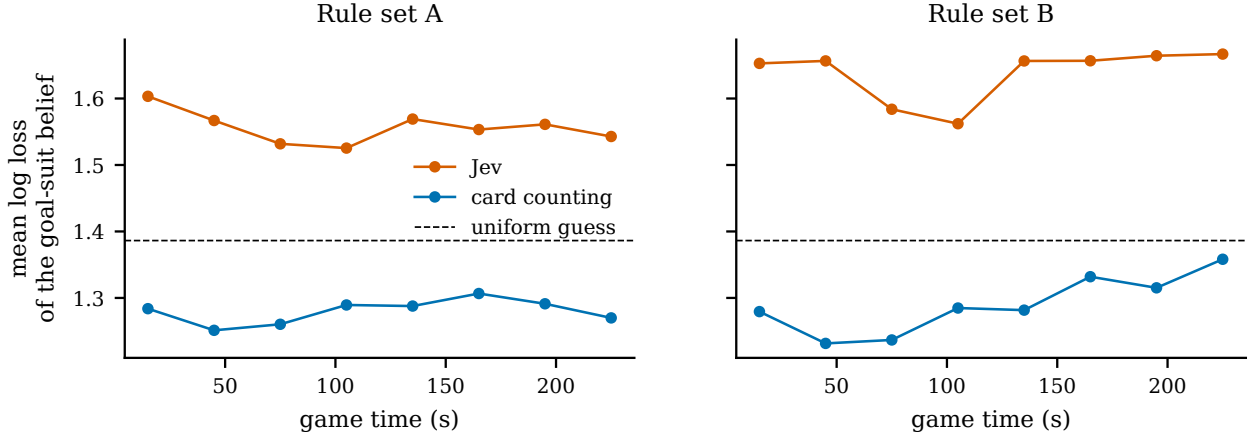


Figure 5: Log loss of the goal-suit belief over game time (30-second bins), all Jev decisions in all conditions of a rule set. Card counting is computed at the same moments from the same public information and Jev's own hand. Lower is better; the dashed line is a uniform guess.

How Jev trades (secondary; Table 8 in the appendix).

- **Jev trades about twice as often as its twin:** 26 vs 12 trades per game in its seat in A, and 19 vs 9 in B. The whole market trades more when Jev is in it (41 vs 30 trades per game in A, 26 vs 18 in B).
- **Prices move further from the true value when Jev plays.** The mean distance between the trade price and the true card value (10 for a goal card, 0 for other cards) is 7.3 with Jev and 5.9 with the twin in A, and 9.5 vs 6.1 in B.
- **In B, the market rejects the twin's orders, not Jev's.** The twin has 17 rejected orders per game in B, because its orders do not beat the best bid or ask. Jev has less than 1: its menu lists only prices that the rules accept, and the few rejections come from quotes that change during the 0.3-second delay.

Belief in full games. Jev's goal-suit belief in games is as poor as in stage 1: the mean log loss over all decisions is 1.55 in A and 1.63 in B, against 1.27 and 1.28 for card counting at the same moments and 1.386 for a uniform guess. Figure 5 shows that Jev's belief does not improve during the game, while the game reveals more information with every trade. H1c is not supported in full games either.

5 Discussion

Summary of the hypotheses.

Hypothesis	Result	Evidence
H1a Jev’s belief not worse than card counting	not supported	log loss +0.17 to +0.28 (st. 1), +0.28 to +0.35 (st. 2)
H1b card counting in state: too confident	supported	top probability 0.69 vs 0.49 given
H1c belief improves later in the game	not supported	Figure 5
H2a paper’s ranking in A	partly	holds in 240-s games, not in 10,000-event games
H2b B: fewer trades, smaller fundamentalist edge	supported	35 vs 114 trades; 112 vs 222 chips
H2c Jev – twin differs between A and B	supported	A – B = +24 [11, 36]
H3a simple rules obeyed	partly	“never take” broken 1–14% of the time
H3b calculations not done	supported	fundamentalist overlap 0.36–0.43
H3c words act literally	supported	behaviour noise trader buys 57% vs 8%
H3d wording effect same in A and B	not supported	algorithm better in 5/6 types in A, 2/6 in B
H4a neutral Jev between f and n	partly	below f; below n in A, not clear in B
H4b similar decisions, similar profit	not supported	$\rho(\text{overlap}, d) = +0.52$

What the results mean. A persona text changes what Jev does. It does not make Jev trade like the trader that the text describes, and it does not make Jev earn what that trader earns. Three results explain this.

1. **Jev does not know the goal suit.** In both stages, Jev’s belief is worse than a uniform guess, and it does not improve during the game. The fundamentalist, the only rule-based trader that earns money, depends on this belief. So the fundamentalist persona is the one that fails most (84 to 105 chips per game behind its twin).
2. **Jev follows the words, not the procedure.** Jev obeys short rules well and reads descriptions literally. It does not carry out a calculation that the text describes. A persona is therefore a useful way to steer Jev’s *style* (how often it acts, which side it takes), but not its *judgement*.
3. **Jev spreads its probability.** We sample from Jev’s probabilities over up to 253 options. Jev puts some probability on many prices, while each twin chooses one price by rule. Sampling therefore makes Jev trade often and at prices far from the true value. We think that this is the main reason why Jev trades twice as often as its twin and why market prices become less accurate when Jev plays (we did not test this directly).

Market rules. The rules change the outcome as much as the persona does. The regression effect of rule set A (+18 chips) is larger than any wording effect. In B, every trade clears all quotes, and only an order that beats the best quote is accepted. The rule-based twins lose many orders to this rule (17 rejected orders per game); Jev does not, because its menu has only accepted prices. But Jev still earns less in B. In B, the prices of trades with Jev are further from the true value (9.5 vs 6.1 with the twin; 7.3 vs 5.9 in A), and Jev’s belief is worse (log loss 1.63 vs 1.55). We think that these two facts cause the larger loss in B, but our design does not test this. The rule sets are a design choice with a large effect, and a study of AI traders should report more than one.

For simulation of market participants. Our goal was to simulate real market participants with Jev. The results say: with a text persona only, Jev gives a trader with its own recognisable style, not a copy of a given model trader. It is closest to the passive types (market makers, noise trader) and furthest from the informed type (fundamentalist). If a simulation needs informed traders, the information must be part of the design; it cannot come from the persona text. The +assist result (stage 1) shows that giving Jev the calculated belief is not enough either, because Jev then becomes too confident.

Limitations.

- **Fewer games than planned.** The plan fixed 40 games per condition. The run stopped after 38 games per condition, when the spending limit of the API key was reached. All conditions have the same games (the run goes in waves of one game number for all conditions), so the comparison stays balanced. The

noise check gives a minimum detectable difference of 28–43 chips per condition with 40 games; with 38 games it is about 3% larger.

- **Bottom-feeder persona text.** The bottom-feeder persona says that it copies “each other player”, but the twin copies only the fixed fundamentalist opponent (as in the paper). Jev’s view of the text is therefore wider than the twin’s rule.
- **Homogeneous noise-trader games do not trade.** The paper does not say what a noise trader does in a suit with no bid. In our reading, it does not trade there, so four noise traders never trade, while the paper reports trades. This does not change stage 2, where each game has a fundamentalist and a bottom-feeder.
- **Card counting is a weak reference.** The paper’s calculation uses only the cards seen in trades and the player’s own hand. Its log loss is only slightly below a uniform guess. A stronger reference (for example, a model of the other players’ behaviour) could show an even larger gap to Jev.
- **One model, one decoding.** We test one version of Jev (`typesafe/jev-1.13`) and sample from its full distribution, as fixed in advance. Choosing Jev’s most probable option instead would give different, less random behaviour (in an earlier pilot, argmax decoding made Jev pass almost always).
- **Values read from figures.** The paper’s values in Figure 1 are read by eye from its figures.

6 Reproduction

All code is in the repository (branch `claude/project-thread-vvzuh9`). Each output directory has a `run.json` with the code version and all settings; each game record has the seed, the deal, every order, trade and cancel, and for Jev seats each decision with its probability and goal-suit belief; each Jev request is logged with its game, seat and decision number.

```
python -m figgie.experiments.replicate --out results/final/stage0
S1=results/final/stage1
python -m figgie.experiments.stage1 moments --out $S1/moments.jsonl
for s in 1a 1b 1c; do
    python -m figgie.experiments.stage1 ask --study $s --backend jev \
        --moments $S1/moments.jsonl --out $S1/$s.jsonl
done
python -m figgie.experiments.stage1 analyse --moments $S1/moments.jsonl \
    --rows $S1/1a.jsonl $S1/1b.jsonl $S1/1c.jsonl --out $S1/report.json
python -m figgie.experiments.sweep --backend jev --games 40 --out results/final/stage2
python -m figgie.experiments.paired --run results/final/stage2 --stage1 $S1/report.json
python paper/final/make_figures.py
```

References

- [1] A. Ozerov, S. DiSilvio, Y. Luo. Traders in a Strange Land: Agent-based discrete-event market simulation of the Figgie card game. *arXiv:2110.00879*, 2021.
- [2] Jane Street. Figgie. <https://www.janestreet.com/figgie/>.
- [3] C. Chiarella, G. Iori, J. Perelló. The impact of heterogeneous trading rules on the limit order book and order flows. *J. Economic Dynamics and Control* 33(3):525–537, 2009.
- [4] L. Glosten, P. Milgrom. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *J. Financial Economics* 14(1):71–100, 1985.
- [5] M. Avellaneda, S. Stoikov. High-frequency trading in a limit order book. *Quantitative Finance* 8(3):217–224, 2008.
- [6] D. Gode, S. Sunder. Allocative efficiency of markets with zero-intelligence traders. *J. Political Economy* 101(1):119–137, 1993.
- [7] D. Byrd, M. Hybinette, T. Balch. ABIDES: Towards high-fidelity multi-agent market simulation. *ACM SIGSIM PADS*, 2020.

- [8] J. Horton. Large language models as simulated economic agents: What can we learn from Homo silicus? NBER Working Paper 31122, 2023.

A Tables

Setting	Line-up (4 of one type)	SD wealth	SD cash	SD payout	Trades per game
A-10k-paper	Noise	50.0	0.0	50.0	0
A-10k-paper	Fundamentalists	57.8	30.0	67.8	26
A-10k-paper	Bottom-feeder	50.0	0.0	50.0	0
A-240s-paper	Noise	50.0	0.0	50.0	0
A-240s-paper	Fundamentalists	57.4	27.8	66.8	25
A-240s-paper	Bottom-feeder	50.0	0.0	50.0	0
B-240s-paper	Noise	50.0	0.0	50.0	0
B-240s-paper	Fundamentalists	57.0	27.4	67.4	21
B-240s-paper	Bottom-feeder	50.0	0.0	50.0	0
A-240s-equal	Noise	50.0	0.0	50.0	0
A-240s-equal	Fundamentalists	57.9	25.9	65.5	22
A-240s-equal	Bottom-feeder	50.0	0.0	50.0	0
B-240s-equal	Noise	50.0	0.0	50.0	0
B-240s-equal	Fundamentalists	56.5	22.8	65.1	16
B-240s-equal	Bottom-feeder	50.0	0.0	50.0	0
Paper	Noise	201.0	188.5	61.4	–
Paper	Fundamentalists	61.8	33.1	68.6	–
Paper	Bottom-feeder	48.6	0	48.6	–

Table 4: Homogeneous games, as the paper’s table 2 (100 games each; SDs over games, averaged over seats).

Rule set	Type in the tested seat	Mean profit	SD profit	SD of difference to fundamentalist
A	Fund.	39	85	–
A	Bottom-f.	-3	52	89
A	Chartist	-23	52	91
A	Noise	-38	35	97
A	MM (GM)	-11	32	80
A	MM (AS)	-11	48	96
B	Fund.	25	70	–
B	Bottom-f.	-4	47	76
B	Chartist	-18	66	82
B	Noise	-33	34	72
B	MM (GM)	-4	39	62
B	MM (AS)	-28	76	92

Table 5: Noise check: the stage 2 line-up with rule-based traders only, equal speed, 100 games.

Rule set	State	Jev			Card counting	
		Log loss [95%]	Brier	Top-1	Log loss	Top-1
A	basic	1.560 [1.42, 1.72]	0.839	0.21	1.296	0.44
A	+summary	1.511 [1.38, 1.66]	0.814	0.22	1.296	0.44
A	+log+summary	1.468 [1.35, 1.60]	0.794	0.26	1.296	0.44
A	+known	1.580 [1.45, 1.72]	0.856	0.15	1.296	0.44
A	+assist	1.951 [1.36, 2.66]	0.857	0.45	1.296	0.44
B	basic	1.652 [1.49, 1.83]	0.891	0.14	1.397	0.41
B	+summary	1.627 [1.46, 1.81]	0.873	0.13	1.397	0.41
B	+log+summary	1.617 [1.45, 1.80]	0.861	0.15	1.397	0.41
B	+known	1.611 [1.45, 1.77]	0.876	0.13	1.397	0.41
B	+assist	1.799 [1.38, 2.23]	0.879	0.37	1.397	0.41

Table 6: Stage 1b: goal-suit belief, 100 moments per rule set. Uniform guess: log loss 1.386, Brier 0.75, top-1 0.25.

Rule set	Type	Overlap with twin			Change vs neutral	
		Algorithm	Behaviour	None	Algorithm	Behaviour
A	Fund.	0.42	0.39	0.32	+0.099*	+0.071*
A	Bottom-f.	0.32	0.36	0.22	+0.095*	+0.137*
A	Chartist	0.38	0.30	0.24	+0.148*	+0.067*
A	Noise	0.55	0.49	0.44	+0.107*	+0.050*
A	MM (GM)	0.04	0.04	0.01	+0.027*	+0.021*
A	MM (AS)	0.05	0.04	0.03	+0.028*	+0.013*
B	Fund.	0.36	0.43	0.39	-0.035*	+0.035*
B	Bottom-f.	0.57	0.47	0.36	+0.211*	+0.112*
B	Chartist	0.42	0.44	0.34	+0.070*	+0.093*
B	Noise	0.55	0.38	0.45	+0.101*	-0.063*
B	MM (GM)	0.06	0.07	0.04	+0.020*	+0.025*
B	MM (AS)	0.10	0.10	0.06	+0.038*	+0.040*

Table 7: Stage 1a: similarity to the twin, 100 moments per cell. *: Holm-corrected $p < 0.05$ (paired bootstrap over games).

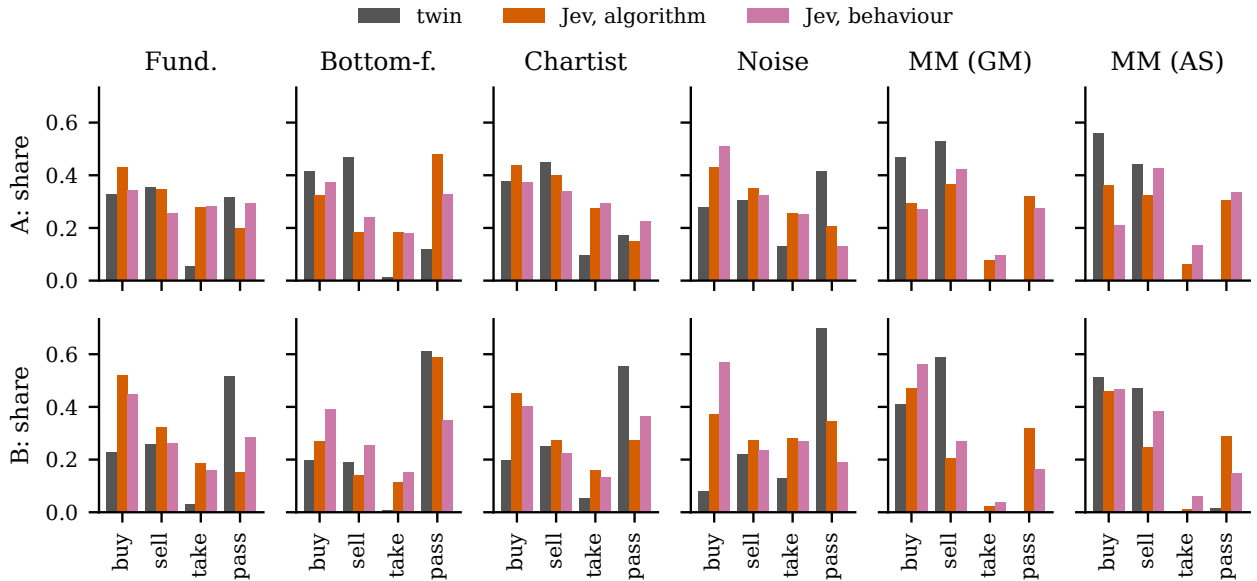


Figure 6: Stage 1a: share of probability on buying (bid or take the ask), selling, taking (either side) and passing.

Rule set	Condition	Seat trades		Passes		Mispricing		Belief log loss	
		Jev	Twin	Jev	Twin	Jev game	Twin game	Jev	Card c.
A	Neutral	34.9	16.1	41	28	8.15	5.99	1.59	1.11
B	Neutral	22.7	11.0	35	20	10.71	5.76	1.63	1.27
A	Fund.	33.4	16.1	27	28	9.50	5.99	1.70	1.20
B	Fund.	24.9	11.0	23	20	10.65	5.76	1.64	1.27
A	Fund. (beh.)	29.9	16.1	48	28	7.25	5.99	1.58	1.28
B	Fund. (beh.)	20.6	11.0	38	20	9.75	5.76	1.58	1.31
A	Bottom-f.	19.0	6.2	74	43	6.52	6.01	1.58	1.29
B	Bottom-f.	11.6	3.2	83	54	9.14	6.22	1.59	1.28
A	Bottom-f. (beh.)	22.9	6.2	47	43	6.59	6.01	1.48	1.34
B	Bottom-f. (beh.)	16.5	3.2	50	54	8.26	6.22	1.64	1.19
A	Chartist	19.6	6.0	57	81	6.65	6.29	1.54	1.33
B	Chartist	12.9	1.9	73	101	9.05	6.89	1.55	1.28
A	Chartist (beh.)	24.7	6.0	49	81	9.56	6.29	1.64	1.41
B	Chartist (beh.)	17.4	1.9	57	101	12.51	6.89	1.70	1.42
A	Noise	32.8	24.3	25	46	7.50	5.90	1.48	1.38
B	Noise	25.6	11.2	54	69	9.81	6.28	1.71	1.34
A	Noise (beh.)	37.0	24.3	20	46	8.07	5.90	1.57	1.28
B	Noise (beh.)	26.5	11.2	32	69	9.68	6.28	1.66	1.28
A	MM (GM)	20.2	8.2	42	66	5.74	5.18	1.50	1.23
B	MM (GM)	15.2	13.1	37	44	9.28	4.81	1.62	1.25
A	MM (GM) (beh.)	22.8	8.2	41	66	6.04	5.18	1.53	1.23
B	MM (GM) (beh.)	17.3	13.1	23	44	8.71	4.81	1.69	1.30
A	MM (AS)	20.0	9.8	37	32	6.68	6.02	1.51	1.25
B	MM (AS)	13.5	10.9	34	39	7.75	6.75	1.63	1.26
A	MM (AS) (beh.)	21.8	9.8	42	32	6.04	6.02	1.49	1.20
B	MM (AS) (beh.)	18.4	10.9	24	39	8.30	6.75	1.57	1.21

Table 8: Stage 2, secondary measures (means per game, 38 games per cell). Seat trades: trades of the tested seat. Passes: decisions with no order. Mispricing: mean $|\text{price} - \text{value}|$ over all trades of the game, with value 10 for a goal card and 0 for other cards. Belief: mean log loss of Jev’s goal-suit belief over its decisions, and of card counting at the same moments (uniform guess: 1.386).